ICC-1 Cardiology: high yield

Year 3, 2026-2027. One page per lecture, then one summary per Specification Grid topic.

Part 1: lectures

1. ECG interpretation and reporting [070, 086]
2. ECG workshop and ECG revision [086]
3. Tachyarrhythmias and bradyarrhythmias [078, 079] (lecture and LGW)
4. Valvular heart disease and heart murmurs [073]
5. Infective endocarditis [019] (lecture and LGW)
6. Heart failure [072]
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10. Acute coronary syndrome LGW [028, 064]
11. Hypertension [090]
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13. Acute heart failure [028, 061] (lecture and LGW)
14. Aortic dissection [076]
15. Chest pain and cardiac presentations [048, 055]
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Part 2: topic summaries

1. Arrhythmias and ECG interpretation
2. Valvular heart disease and infective endocarditis
3. Chronic heart failure, cardiomyopathies and pericardial disease
4. Ischaemic heart disease, dyslipidaemia and hypertension
5. Cardiac emergencies (acute heart failure, aortic dissection)
6. Cardiac symptoms, presentations and clinical reasoning

PART 1: LECTURE

ECG interpretation and reporting [070, 086]

Topic: Arrhythmias and ECG interpretation

Leads and paper

- 10 electrodes (4 limb, 6 chest) give 12 leads; right leg = ground. V4 = 5th ICS midclavicular; V1/V2 = 4th ICS right/left sternal edge.
- Small square 0.04 s; large square 0.2 s. Check settings first: 25 mm/s, 10 mm = 1 mV, aVR negative.

Normal values and axis

- PR 120-200 ms (3-5 small squares); QRS < 110 ms; QTc long > 440 ms (men) / > 460 ms (women); > 500 ms = torsades risk.
- Transition V3-V4. At/before V2 = counterclockwise (RV enlargement); after V4 = poor R progression (COPD, old anteroseptal MI).
- Normal axis -30 to +90. Lead I +, III - = LAD (conduction abnormalities); I -, III + = RAD (RVH); both - = extreme.

Ischaemia and infarction (not with LVH or BBB)

- STE in 2 contiguous leads ≥ 1 mm; in V2-V3 ≥ 2 mm men ≥ 40, ≥ 2.5 mm men < 40, ≥ 1.5 mm women. ST depression ≥ 0.5 mm.
- Pathological Q: > 1 mm wide and > 2 mm deep, or > 25% of the next R.
- Age: 0-6 h STE/hyperacute T; 6-24 h Q + STE; 24 h-3 wk Q, ST at baseline, inverted T; > 3 wk Q only. STE > 3 wk, no pain, normal enzymes = LV aneurysm.
- Anteroseptal V1-V4; anterolateral V3-V6, I, aVL; posterior = tall R, ST depression, tall T in V1-V3; RV = STE V3R-V4R. Inferior MI: reciprocal change in lateral leads.

Chambers and others

- LAE: broad notched P in II, deep negative P in V1. RAE: tall peaked P > 2.5 mm in II.
- LVH: S V1 + R V5 > 35 mm (or R aVL ≥ 11 mm); strain V5-V6. RVH: R in V1 > 7 mm, deep S V6; strain V1-V2.
- Low voltage + electrical alternans = massive pericardial effusion.
- U waves = low K; long QT = low K, Ca, Mg; short QT = high Ca; sagging ST = digitalis.
- Pacing spike before P = atrial, before QRS = ventricular, both = dual chamber.

Source: ECG interpretation [014.1]; ECG reporting [016.1]

PART 1: LECTURE

ECG workshop and ECG revision [086]

Topic: Arrhythmias and ECG interpretation

Chest pain: life-threatening four

- Always exclude acute MI, aortic dissection, pulmonary embolism and tension pneumothorax.
- Check BP and pulse volume in both arms before antithrombotics: antiplatelets, anticoagulants and thrombolytics are absolutely contraindicated in dissection.
- STEMI: aspirin 300 mg + clopidogrel 300 mg, analgesia, nitrates, urgent primary PCI (preferred) or thrombolysis if pain < 12 h. Do not wait for enzymes: minutes mean muscle.
- Non-ST elevation ACS: about 50% have a normal or non-diagnostic ECG; diagnose on typical pain plus risk factors. Diffuse ST depression with STE in aVR is managed as NSTE-ACS (urgent angiography).
- Treat NSTE-ACS with aspirin, clopidogrel, nitrates, beta-blocker, enoxaparin 1 mg/kg 12-hourly, CCU, enzymes, echo.

Classic cases

- PE (OCP, unilateral leg swelling): sinus tachycardia, RBBB, RAD, S1Q3T3. Confirm with CT pulmonary angiography; anticoagulate.
- Aortic dissection: tearing pain to the back, unequal pulses and BP, early diastolic murmur (aortic regurgitation); ECG may show LVH with strain. Echo and CT aortography, then urgent surgery.
- Pericarditis: pleuritic pain after viral illness, rub; widespread concave STE with PR depression, reciprocal changes in aVR and V1. NSAIDs + colchicine.
- Tamponade: Beck's triad (hypotension, raised JVP, muffled sounds); echo shows RV collapse. Urgent pericardiocentesis.
- Hyperkalaemia (ESRD): peaked T, long PR, broad bizarre QRS, sine wave. IV calcium over 10 min, glucose-insulin, bicarbonate, urgent dialysis.

Rhythm cases

- Wide-complex tachycardia with structural heart disease = VT until proved otherwise: unstable, synchronised DC shock; stable, IV amiodarone.
- Regular narrow tachycardia ~200, no P = SVT: IV verapamil or beta-blocker. Irregular narrow, no P = AF: rate control + anticoagulation.
- Complete heart block: regular, variable PR, slow (~35): atropine, then pacemaker.
- Heart block: uniform long PR, all conducted = 1st degree; uniform PR with dropped beats = Mobitz II; variable PR, irregular = Mobitz I.

Conduction and pre-excitation

- LAFB: LAD, qR in I/aVL, rS in inferior leads. LPFB: RAD, rS in I/aVL, qR inferior. Bifascicular = RBBB + hemiblock; trifascicular = bifascicular + 1st degree block.
- WPW: short PR, delta wave, wide QRS. Dominant R in V1 = type A (left pathway); dominant S in V1 = type B (right pathway).

Source: ECG workshop [015.2]; ECG revision [017.5]

PART 1: LECTURE

Tachyarrhythmias and bradyarrhythmias [078, 079] (lecture and LGW)

Topic: Arrhythmias and ECG interpretation

Identify the pacemaker

- Narrow QRS < 120 ms = supraventricular; wide ≥ 120 ms = ventricular. Rate: regular = 300 / large squares in R-R; irregular = QRS in 30 large squares x 10.
- Sinus: P upright in II, negative in aVR. Junctional: inverted P in II, upright in aVR.
- Sinus arrhythmia: rate varies with breathing (R-R difference ≥ 120 ms). MAT: ≥ 3 P shapes, rate > 100 (< 100 = wandering pacemaker).
- VT: wide, regular, AV dissociation, T opposite QRS. Torsade: twisting polymorphic VT with long QT. VF: chaotic, no P/QRS/T.

Atrial fibrillation

- Pulmonary vein foci (300-600/min); no atrial contraction, so LA appendage thrombus and stroke.
- Paroxysmal < 7 days; persistent > 7 days; long-standing > 1 year; permanent = rhythm attempts stopped.
- Unstable: 5000 IU UFH then synchronised DC shock. Stable: rate control first-line (or rhythm) + OAC.
- LVEF < 40%: no verapamil/diltiazem (beta-blocker or digitalis). Structural disease: no flecainide/ propafenone (amiodarone).
- Before cardioversion: TEE to exclude thrombus or 3 weeks OAC; OAC for 4 weeks after.
- Warfarin for metallic valve or mitral stenosis regardless of score. Otherwise CHA2DS2-VA 2 = OAC, 1 = consider; DOACs preferred. No antiplatelets; LAA occlusion if OAC contraindicated.

Other tachycardias

- SVT: unstable = synchronised DC; stable = IV adenosine, beta-blocker or verapamil/diltiazem.
- VT: unstable = synchronised DC; stable = IV amiodarone; long term amiodarone or ICD.

Heart block and bundle branch block

- 1st degree: PR > 200 ms, all P conducted. Mobitz I: PR lengthens until a drop (AV node; monitor). Mobitz II: fixed PR, sudden drop, wide QRS (infranodal; pace).
- Complete block: narrow escape 50-60 (His, reliable); broad escape 15-40 (His-Purkinje, Stokes-Adams; permanent pacemaker).
- Symptomatic block: correct cause, IV atropine, temporary then permanent pacing.
- LBBB: QS/rS in V1, broad notched R in I, V5-V6. RBBB: RSR' in V1, slurred S in V6.
- WPW: short PR, delta wave; dominant R in V1 = type A (left), dominant S = type B (right).

Source: Tachy-Bradyarrhythmias lecture [018.1/019.1]; LGW Parts 1 and 2 [020.2]

PART 1: LECTURE

Valvular heart disease and heart murmurs [073]

Topic: Valvular heart disease and infective endocarditis

Aortic stenosis (AS)

- Causes: senile calcification, bicuspid valve, rheumatic. Asymptomatic for years, then exertional syncope, angina, dyspnoea; untreated death in 2-3 years once symptoms start.
- Slow-rising small pulse; sustained, non-displaced apex; harsh diamond-shaped ejection systolic murmur at aortic area radiating to carotids.
- Severity: late-peaking murmur, soft/absent A2, reversed S2 splitting. Loudness is NOT a guide.
- CXR: small heart, post-stenotic dilated ascending aorta. ECG: LVH with strain.
- Valve replacement if severe + symptoms; if asymptomatic, operate for EF <50%, symptoms/ hypotension on exercise test, peak velocity >5.5 m/s. TAVI if unfit for surgery.

Aortic regurgitation (AR)

- Collapsing pulse, De Musset, Quincke, Duroziez (severe AR), pistol-shot femorals; apex displaced down and out.
- Soft early diastolic murmur at left sternal edge, leaning forward in expiration; Austin Flint = apical mid-diastolic rumble from the jet hitting the anterior mitral cusp.
- Surgery: acute severe AR (e.g. endocarditis), symptoms, EF ≤50%, or LV dilated (EDD >70 mm / ESD >50 mm).
- Tissue valves for elderly or when anticoagulation must be avoided; not for young (rapid degeneration).

Mitral stenosis (MS)

- Almost always rheumatic. Dyspnoea, haemoptysis, malar flush, AF with emboli.
- Tapping apex, loud S1, opening snap, low-pitched mid-diastolic rumble: bell at apex, left lateral, expiration. Graham Steell = PR from pulmonary hypertension.
- CXR: LA enlargement, double shadow. ECG: P mitrale, AF.
- AF: rate control or cardioversion PLUS anticoagulation. Intervene if area ≤1.5 cm^2 with symptoms or pulmonary hypertension.
- Balloon valvuloplasty: TOE first to exclude LA thrombus; contraindicated with heavy calcification, more than mild MR, LA thrombus.

Mitral regurgitation and right-sided lesions

- MR: soft S1, pansystolic murmur apex to axilla, S3. MVP: mid-systolic click then late systolic murmur.
- Surgery: symptomatic severe MR, or asymptomatic with ESD >40 mm or EF <60%. Papillary/ chordal rupture = emergency valve surgery.
- TR: large cv waves, pulsatile liver, pansystolic murmur at lower left sternal edge louder on inspiration; usually functional (RV dilatation).

Auscultation tips

- Time murmurs with the carotid pulse: starts with it = systolic; starts after it = diastolic.

Source: [073] Valvular heart disease; [073] Heart murmurs auscultation LGW

PART 1: LECTURE

Infective endocarditis [019] (lecture and LGW)

Topic: Valvular heart disease and infective endocarditis

Pathogenesis and risk

- Needs two things: damaged endocardium (turbulent flow → platelet-fibrin clot) plus bacteraemia; bacteria adhere and multiply to form vegetations that embolise and trigger immune complexes.
- Risks: prosthetic valves, prior IE, congenital and rheumatic heart disease, degenerative aortic valve, severe MVP, IV drug use, intravascular devices, invasive procedures. Can affect normal valves.
- People who inject drugs: tricuspid valve in about 50%; right-sided lesions give septic pulmonary emboli.

Organisms

- Native valve: viridans streptococci most common (54%), then S. aureus (19%), enterococci.
- Early prosthetic (<6 months): S. aureus, coagulase-negative staph (S. epidermidis). Late prosthetic: same as native.
- Culture negative (up to 20%): prior antibiotics, fastidious strep, HACEK, Coxiella, Candida; use prolonged/fungal cultures and paired serology.

Clinical features

- Fever in >95%, fatigue, breathlessness; acute presentation usually heart failure.
- Signs: splinter haemorrhages, Osler nodes, Janeway lesions, conjunctival haemorrhages, Roth spots, splenomegaly.

Diagnosis

- Blood cultures are the key test: at least 2, ideally 3 sets BEFORE antibiotics, 8-10 mL per bottle.
- Echo: TTE then TOE. Also FBC, U&E, CRP, urine microscopy, CXR.
- 2023 ESC criteria. Major: typical organism in 2 separate cultures; positive imaging. Minor: predisposition, fever >38 °C, embolic vascular (Janeway, conjunctival haemorrhage), immunological (Osler, Roth, GN, RF), microbiology.
- Definite: 2 major, 1 major + 3 minor, or 5 minor. Possible: 1 major + 1-2 minor, or 3-4 minor.

Treatment

- Empirical IV high dose after cultures: native = amoxicillin + gentamicin; prosthetic = vancomycin + gentamicin; then tailor to organism and MIC.
- Surgery: heart failure from valve damage, uncontrolled infection (abscess, heart block, fungal/resistant organism), embolic risk (large vegetation, recurrent emboli).
- Remove infected lines/devices; dental review.
- Prophylaxis for high-risk dental procedures: single dose 30-60 min before, e.g. amoxicillin 2 g orally (adults).

Source: [019] Infective Endocarditis simplified; [019] IE LGW Answers

PART 1: LECTURE

Heart failure [072]

Topic: Chronic heart failure, cardiomyopathies and pericardial disease

Definition and pathophysiology

- Clinical syndrome of cardinal symptoms (breathlessness, ankle swelling, fatigue) ± signs, from structural/functional heart abnormality raising intracardiac pressures and/or lowering cardiac output.
- CO = SV × HR. Myocardial loss activates SNS and RAAS (harmful remodelling) and natriuretic peptides (protective, from stretch).
- Treatment aim: block SNS and RAAS, enhance natriuretic peptides.
- Causes: myocardial (CAD, hypertension, cardiomyopathy), valvular, pericardial, arrhythmia, high-output states (anaemia, thyrotoxicosis, Paget's, AV fistula, sepsis), volume overload.

Clinical features

- Left-sided: dyspnoea, orthopnoea, PND (most specific), bibasal fine crackles.
- Right-sided: leg oedema, RUQ pain, raised JVP, hepatojugular reflux, ascites. Cor pulmonale = RV failure from lung disease.
- Apex: LV enlargement = downward and outward, localised; RV = outward, diffuse, systolic retraction.
- S3 (just after S2): volume overload; normal in children, pregnancy, athletes. S4 (just before S1): stiff ventricle (hypertension, CAD); not normal even in elderly.

Classification

- LVEF: HFrEF ≤40%, HFmrEF 41-49%, HFpEF ≥50%.
- NYHA I none; II ordinary activity; III less-than-ordinary activity; IV symptoms at rest.
- ACC/AHA: A at risk; B structural disease, no symptoms; C symptomatic; D advanced.

Investigations

- Natriuretic peptides: normal level in an untreated patient virtually excludes HF. Rule-out: BNP <100, NT-proBNP <300 pg/mL.
- Echo is the key test of cardiac function. Normal ECG makes HF unlikely. CXR finds other causes of breathlessness.
- Iron studies (ferritin, transferrin saturation): about a third are iron deficient.
- Cardiac MRI: poor echo windows; infiltrative disease (amyloid, sarcoid), myocarditis.

Treatment

- Prevention first; lifestyle: salt restriction, exercise, stop smoking/alcohol, weight.
- Diuretics improve symptoms only.
- Four prognostic pillars: ACEI/ARB/ARNI, MRA, beta-blocker, SGLT2 inhibitor.
- Sacubitril inhibits neprilysin, raising natriuretic peptides. SGLT2i increase distal sodium delivery, lowering preload and afterload.
- Devices/surgery: revascularisation, valve replacement, biventricular pacemaker, LVAD, transplant.

Source: [072] Heart Failure, Dr Mina Samy

PART 1: LECTURE

Cardiomyopathies [063]

Topic: Chronic heart failure, cardiomyopathies and pericardial disease

Classification

- Myocardial diseases affecting mechanical or electrical function. ESC groups: HCM, arrhythmogenic (ACM), DCM, restrictive (RCM), unclassified (LVNC, Tako-tsubo); each familial or non-familial.
- DCM causes mnemonic APCDEF MN: Alcohol, Pregnancy, Calcium/phosphate low, Drugs (anthracycline), Endocrine (thyroid), Familial, Fast HR, Myocarditis, Nutritional (thiamine, carnitine, selenium).

Hypertrophic cardiomyopathy (HCM)

- Inherited (mostly autosomal dominant, sarcomere genes MYH7, MYBPC3) hypertrophy without AS or hypertension; 1 in 500; commonest cause of sudden death in the young.
- Asymmetric septal hypertrophy; ~25% have dynamic LVOT obstruction (SAM of anterior mitral leaflet). AF causes rapid deterioration/pulmonary oedema.
- Signs: double apical impulse (S4), jerky carotid pulse, ejection murmur louder with standing/Valsalva, softer with squatting; MR pansystolic murmur.
- ECG: LVH, ST-T changes, inferolateral Q waves. Echo usually diagnostic; CMR shows fibrosis and separates HCM from infiltration.
- SCD risk factors: wall >30 mm, family SCD <50 y, NSVT on Holter, unexplained syncope, flat/hypotensive BP response on exercise. Two or more: ICD; lower risk: amiodarone.
- Symptoms: beta-blocker and/or verapamil; disopyramide if LVOT obstruction; dual-chamber pacing, alcohol septal ablation (risk: heart block, MI) or surgical myectomy. Avoid vasodilators.

ARVC

- Fibro-fatty replacement of RV: early VT/sudden death, later RV or biventricular failure.
- ECG: T inversion V1-V3, epsilon waves, RBBB; Holter RV ectopics/NSVT; CMR shows fibro-fat. Task Force Criteria.
- Beta-blocker first line; amiodarone/sotalol if symptomatic; ICD if life-threatening; transplant occasionally.

DCM and acquired forms

- Dilated ventricles, systolic dysfunction, normal wall thickness; HF, arrhythmia, emboli, sudden death. Exclude CAD (angiography/CTCA) if >40 y or risk factors; biopsy not routine. Treat as HF; CRT/ICD in NYHA III-IV; transplant.
- Peripartum: last trimester to 5 months postpartum; obese, multiparous, >30 y, pre-eclampsia; ~half recover in 6 months. Tachycardiomyopathy: cardioversion/ablation allows recovery.
- Tako-tsubo: older women, MI-like, catecholamine excess, normal coronaries, apical ballooning; recovers in 4-6 weeks.

Restrictive and LVNC

- RCM: small/normal ventricles, bi-atrial enlargement, near-normal EF, impaired filling; amyloid commonest. Raised JVP, Kussmaul's sign, S3/S4.
- Amyloid clues: macroglossia, periorbital bruising, neuropathy, carpal tunnel, proteinuria, AF, effusions; voltage-mass mismatch; speckled myocardium on echo.
- Biopsy often useful (unlike other CMs); cath separates from constriction. Treat HF/AF (anticoagulate); tafamidis for familial amyloid; liver transplant in familial type.
- LVNC: spongy trabeculated LV apex; HF, emboli, arrhythmia, sudden death.

Source: ICC-1 [063] Cardiomyopathies 2026-2027

PART 1: LECTURE

Pericardial diseases [063]

Topic: Chronic heart failure, cardiomyopathies and pericardial disease

Pericardium and causes

- Double-layered fibro-serous sac: visceral (epicardium) and parietal (fibrous + serous) layers; normally 15-25 ml fluid; blood supply from internal mammary arteries.
- Causes (RHIM + malignancy): radiotherapy; haemopericardium (trauma, aortic dissection); infection (viral, bacterial, TB, fungal), inflammatory (connective tissue disease, rheumatic fever, Dressler's, drugs), idiopathic; metabolic (uraemia, hypothyroidism); malignancy.

Acute pericarditis

- Pleuritic-type chest pain relieved by leaning forward.
- Friction rub: rough scratching, louder leaning forward, not tied to cardiac cycle, often absent.
- ECG: widespread saddle-shaped ST elevation (not aVR), PR depression, T inversion; low voltage and electrical alternans if large effusion.
- CXR normal unless large effusion (globular heart); echo to exclude significant effusion; bloods guided by likely cause.
- Treatment: NSAID; add colchicine to reduce symptoms and recurrence; steroids if autoimmune or resistant; antimicrobials for bacterial/TB/fungal; dialysis for uraemic; pericardiocentesis if tamponade.

Effusion and tamponade

- Tamponade: raised intrapericardial pressure limits ventricular filling and cardiac output; rapid accumulation raises pressure fast.
- Symptoms: breathlessness, dizziness, syncope, chest pain.
- Signs: tachycardia, hypotension, weak pulses, pulsus paradoxus, absent apex, faint heart sounds, raised JVP, cold peripheries (slide also lists Kussmaul's sign).
- Beck's triad: raised JVP + hypotension + quiet heart sounds.
- Pulsus paradoxus: inspiratory systolic BP fall >10 mmHg (normal <10) from marked septal shift to the left.
- ECG low voltage + electrical alternans; CXR globular heart; echo is the best non-invasive test.
- Bloods: thyroid, ANA, Quantiferon; clotting and platelets before pericardiocentesis. Fluid: cytology, GeneXpert PCR for TB.
- Treatment: emergency percutaneous pericardiocentesis; surgical drainage if not accessible, traumatic or bloody; pleuro-pericardial window for recurrent (e.g. malignant) effusion; treat cause.

Constrictive and adhesive pericarditis

- Constrictive: TB is the commonest cause (per lecture); pericardial knock; CT/MRI most sensitive; main differential is restrictive cardiomyopathy.
- Adhesive: rare, few manifestations; may follow rheumatic fever (adhesion to mediastinum/chest wall).

Source: ICC-1 [063] Pericardial Diseases 2026-2027

PART 1: LECTURE

Coronary artery disease [071]

Topic: Ischaemic heart disease, dyslipidaemia and hypertension

Pathophysiology and risk

- CAD = CHD = IHD: atherosclerotic narrowing of coronaries; leading cause of death worldwide. Stable CAD: fixed 80-90% stenosis; enough flow at rest, ischaemia when demand rises.
- ACS: plaque often only 30-50% but unstable; rupture/erosion exposes vWF, platelet aggregation, thrombus.
- Ischaemia = supply/demand imbalance. Low supply: atherosclerosis, spasm, thrombus/embolus, vasculitis; hypoxia, anaemia, hypotension. High demand: exercise, tachycardia, hypertension, LVH.
- Modifiable: smoking, diabetes, hypertension, dyslipidaemia, inactivity, obesity. Non-modifiable: age, sex, family history, ethnicity.

Coronary anatomy

- LAD ("widow maker", ~60% of LV): anterior septum, anterior and lateral walls, apex.
- LCX: left AV groove; lateral wall (obtuse marginals), posterior wall in 10-20%.
- RCA: right AV groove; RA, RV, posterior LV wall via PDA in 80%. SA node: RCA 65%, LCX 25%. AV node: RCA 90%, LCX 10%.

Presentations

- Typical pain: retrosternal, to jaw/arm/back/epigastrium, triggered by exertion/emotion, relieved by rest/SL nitrate. Angina equivalents: dyspnoea, fatigue, faintness.
- ACS spectrum: UA (troponin negative), NSTEMI (troponin positive; ST depression/T changes/ normal), STEMI (ST elevation or new LBBB).
- Variant (Prinzmetal): rest angina, midnight-early morning, transient ST elevation.
- High-risk stable angina: post-MI angina, poor effort tolerance, ischaemia at low workload, left main/3-vessel disease, poor LV function.

Investigations

- Resting ECG/echo usually normal in stable CAD. Labs: CBC, U&E, lipids, glucose, hs-troponin.
- Pathway: ECG, then CTCA (1st line), then MPS/stress echo/stress MRI (2nd), then invasive angiography (3rd).
- Calcium score 0: obstructive CAD very unlikely (~0.1%/yr); >100: high risk (>2%/yr).
- Thallium perfusion scan: exercise best; dobutamine if unable to exercise.

Treatment

- Nitrates: dilate veins/arteries, cut preload/afterload; headache, flushing, hypotension; nitrate-free period; never with sildenafil.
- Beta-blockers: lower HR, BP, contractility; bradycardia, bronchospasm, cold peripheries, impotence; never stop abruptly (withdrawal syndrome).
- CCBs: oedema, headache, flushing; short-acting nifedipine worsens ischaemia in MI.
- ACEi/ARB: cough, hyperkalaemia, renal impairment. Statins: cramps, raised liver enzymes. Antiplatelets: bleeding, gastritis.
- Nicorandil (K channel activator): no tolerance. Ivabradine (If): slows sinus node, no negative inotropy, safe in HF. Interventional: PCI or CABG.

Source: ICC-1 [071] CAD NGU; CAD Dr Mina Samy 2026-2027

PART 1: LECTURE

Acute coronary syndrome LGW [028, 064]

Topic: Ischaemic heart disease, dyslipidaemia and hypertension

Suspecting ACS

- Check history, risk factors (e.g. smoking), previous IHD, prior tests.
- Suggestive: pain in chest/arms/back/jaw >15 min; pain with nausea, vomiting, sweating, breathlessness or haemodynamic instability; new pain or abrupt worsening of stable angina at little/no exertion.
- If ECG is non-diagnostic: serial ECGs, compare old ECGs, extra leads; consider PE, aortic dissection, pneumonia.

Immediate management

- Pain relief: GTN and/or IV opioid.
- Aspirin 300 mg single loading dose (unless allergic), then continue indefinitely; other antiplatelets only in hospital.
- Resting 12-lead ECG; send to hospital before arrival.
- Oxygen only if SpO2 <94%: target 94-98%; COPD at risk of hypercapnia 88-92% until blood gas.
- Monitor: pain, pulse, BP, rhythm, SpO2, repeated ECGs, pain-relief effect.

Diagnosis

- Troponin repeat 10-12 h after symptom onset (session answer).
- ACS = rise/fall of troponin with at least one value above 99th percentile URL plus ischaemia: symptoms, new ST-T change or LBBB, new Q waves, or new wall motion loss on imaging.

STEMI cases

- Acute chest pain in ED: 12-lead ECG first.
- Inferior STEMI: right coronary artery. Anterior STEMI: LAD.
- Send troponin I, CBC, glucose, urea/creatinine, clotting.
- Antiplatelet loading dose, then urgent angiography and angioplasty (PCI).

Other chest pain cases

- Prinzmetal (vasospastic): rest/night pain, transient ST elevation. Confirm ischaemia and risk stratify (echo, stress ECG, CTCA, labs). Treat: CCB, nitroglycerine, statin, stop smoking.
- Effort angina (chronic coronary syndrome): labs, stress ECG, echo, CTCA. Treat: aspirin, beta-blocker, ACEi if echo abnormal, nitrate or K-channel opener, statin, lifestyle. Angiography/stenting only if severe ischaemia or medical failure.
- Chest wall inflammatory pain: hours long, worse with movement/cough, tender, normal ECG. CBC, CXR, echo; reassure, anti-inflammatory + antacid, review in 1 week.

Source: ICC-1 [028,064] ACS LGW tutor and student notes 2026-2027

PART 1: LECTURE

Hypertension [090]

Topic: Ischaemic heart disease, dyslipidaemia and hypertension

Definitions (ESC 2024)

- Office BP: non-elevated <120/70; elevated 120-139/70-89; hypertension ≥140/90 mmHg (persistent).
- Out-of-office thresholds: home ≥135/85; ABPM daytime ≥135/85, night ≥120/70, 24-h ≥130/80.
- Primary 90% (gradual, >40 y); secondary 10% (often <40 y, abrupt/severe, potentially curable).
- Resistant HTN: above target on ≥3 classes incl. a diuretic, at maximal tolerated doses.
- Isolated systolic HTN (age >50, arterial stiffness): treat as usual but keep DBP ≥60 mmHg (coronary flow). Consider aortic regurgitation.
- Orthostatic hypotension: fall >20 systolic or >10 diastolic within 1-3 min of standing; use standing BP to guide treatment.

Measurement and diagnosis

- Confirm with out-of-office BP (HBPM/ABPM preferred); both detect white-coat and masked HTN.
- Both arms at first visit, use the higher arm; palpate pulse for AF; manual auscultation in AF and pregnancy.
- Screening >160/100: re-evaluate within days to weeks (≤1 month). >180/110: exclude hypertensive emergency.

Secondary causes and work-up

- Suspect if: <40 y with grade 2+, acute worsening, grade 3/emergency, resistant, extensive organ damage, endocrine or OSA features.
- Primary aldosteronism → aldosterone-to-renin ratio (hypokalaemia). Renovascular → renal Doppler/CT/MR angiography (late onset, flash pulmonary oedema, abdominal bruit). Phaeo → metanephrines. OSA → polysomnography. Cushing's → urinary cortisol/dexamethasone test. Coarctation → echo/aortic CTA.
- Organ damage: fundoscopy, ECG/echo (LVH), creatinine, urine albumin-creatinine ratio.
- Already high risk (no SCORE2 needed): ASCVD/HF, organ damage, CKD, DM, familial hypercholesterolaemia. Otherwise SCORE2 (40-69) / SCORE2-OP (≥70); ≥10% = increased risk.

Treatment

- Hypertension ≥140/90: drugs immediately + lifestyle; start low-dose single-pill dual: ACEi/ARB + CCB or thiazide.
- Uncontrolled at 1-3 months → triple (ACEi/ARB + CCB + diuretic) → add spironolactone.
- Elevated BP: lifestyle; drugs only if high risk and BP ≥130/80 after 3 months; start with monotherapy.
- Monotherapy also for age ≥85, frailty, symptomatic orthostatic hypotension.
- Beta-blockers only with compelling indication (arrhythmia, CAD, HF, pregnancy).
- Targets: SBP 120-129, DBP 70-79 (pregnancy: not <80); lenient <140 in elderly/frail.
- Lifestyle: sodium ~2 g/day, ≥150 min/week exercise, BMI 20-25, DASH/Mediterranean diet, alcohol <100 g/week, stop smoking.

Source: ICC-1 [090b] Hypertension 2026-2027

PART 1: LECTURE

Dyslipidaemia LGW [029]

Topic: Ischaemic heart disease, dyslipidaemia and hypertension

Causes and risk

- Familial hypercholesterolaemia (FH): common autosomal dominant disorder; very high LDL, tendon xanthomas, premature CHD; test siblings and children.
- Drug causes: antiretrovirals, beta-blockers, oral contraceptives, corticosteroids, thiazides.
- LDL-C and non-HDL-C (TC - HDL) track CVD risk; HDL protects; chylomicrons raise TG but not CVD risk.
- TG >500 mg/dL → risk of acute pancreatitis (an emergency).

Measurement and screening

- Fasting affects only TG; fast if TG >400 mg/dL or FH.
- Friedewald (TG <400): LDL-C = TC - (HDL + TG/5).
- After MI lipids fall: measure within 24 h or 1 month later.
- Opportunistic screening: family history of CHD or lipid disorder, xanthoma/xanthelasma, arcus <40 y, DM, HTN, obesity, pancreatitis, dialysis.

Risk categories and targets

- SCORE2 (40-69 y) / SCORE2-OP (70-89 y): age, sex, smoking, TC, HDL, SBP.
- Not used in ASCVD (very high), CKD, DM, FH, LDL >190.
- Extreme risk: recurrent event on maximal statin, or polyvascular disease → LDL <40.
- LDL goals: low <116; moderate <100; high <70 + ≥50% fall; very high <55 + ≥50% fall.
- Non-HDL goals: moderate <130; high <100; very high <85.

Management

- Step I maximal statin → II + ezetimibe or bempedoic acid → III triple (add PCSK9 mAb) → IV quadruple.
- Post-MI on moderate statin: switch to high intensity (atorvastatin 40-80 mg) first.
- Statin intolerance: bempedoic acid (low muscle risk), other statin, lower dose + ezetimibe, alternate-day/weekly.
- PCSK9 mAbs (alirocumab, evolocumab) only for very high risk; inclisiran = siRNA.
- Statin monitoring: baseline lipids, liver enzymes, CK; lipids 3-monthly to target then yearly; stop if ALT >3× or CK >10× normal.
- TG <500 on statin: icosapent ethyl preferred (TG 135-499); fenofibrate if >200.
- TG ≥500: diabetes control, no alcohol, caloric restriction, fibrates/omega-3, volanesorsen (FCS >750), plasmapheresis.

Source: ICC-1 [029] Dyslipidemia 2026-2027

PART 1: LECTURE

Acute heart failure [028, 061] (lecture and LGW)

Topic: Cardiac emergencies (acute heart failure, aortic dissection)

Definition and causes

- AHF: worsening signs and symptoms as the heart cannot pump enough blood for the body's needs at normal filling pressure.
- Causes: ischaemic heart disease, valve disease, AF, hypertension, acute kidney disease (cardiomyopathy in chronic HF).

Pulmonary oedema

- Fluid leaks into interstitium/alveoli; Starling balance of capillary hydrostatic vs plasma oncotic pressure.
- Cardiogenic: sudden rise in pulmonary venous (hydrostatic) pressure. Non-cardiogenic: increased capillary permeability.
- Other types: high-altitude, neurogenic, rapid aspiration of massive effusion/pneumothorax, after cardiopulmonary bypass.

Clinical picture and diagnosis

- Exertional dyspnoea, orthopnoea, PND, pink frothy sputum, tachypnoea, tachycardia.
- Bilateral crackles, wheeze ("cardiac asthma"), raised JVP, pitting oedema.
- Tests: CBC, renal and liver function, ABG, cardiac enzymes, pro-BNP; ECG for ischaemia; echo (EF, wall motion); CXR (oedema, effusions, cardiomegaly).
- Differentials: COPD, chronic bronchitis, ILD, pneumothorax, PE, anxiety, anaemia (e.g. iron deficiency from menorrhagia).

Treatment

- Aims: stabilise, find cause and precipitant, rapid symptom relief.
- Oxygen if SpO2 <90%; NIPPV cuts intubation, ventilation and mortality.
- IV loop diuretic (furosemide 40 mg) = cornerstone; check response at 2-4 h, then raise dose/ frequency.
- IV nitrates: venodilators, lower preload and LV filling pressure. Nitroprusside: potent arterial dilator, afterload reduction.
- Hypoperfusion or shock (e.g. BP 60/40, no urine): inotropes (dobutamine, dopamine) and/or mechanical support.
- Ultrafiltration: removes isotonic plasma water across a membrane without changing electrolytes.
- Morphine IV 5 mg only for pain if present.
- Long-term survival: ACEi/ARB, beta-blocker, aldosterone antagonist.

Source: ICC-1 [028,061] Acute heart failure; [061] AHF LGW (2026-2027)

PART 1: LECTURE

Aortic dissection [076]

Topic: Cardiac emergencies (acute heart failure, aortic dissection)

Pathophysiology and risk

- Bleeding within the tunica media; an initial intimal tear is found in 90%. The false lumen usually extends anterograde and may re-enter the true lumen or rupture through the adventitia.
- Rare but catastrophic: untreated early mortality up to 1% per hour; peak age about 63; male:female 2:1.
- Raised wall stress: **hypertension (72%)**, cocaine/stimulants, trauma (including coronary intervention), deceleration injury, weight lifting, coarctation.
- Medial abnormality: Marfan, Ehlers-Danlos, Turner, bicuspid aortic valve; vasculitis (giant cell, Takayasu, RA, syphilis); pregnancy, polycystic kidney disease.

Classification and presentation

- Stanford: **type A involves the ascending aorta**; type B does not.
- Sudden severe **tearing** pain, maximal at onset: retrosternal (type A) or interscapular (type B); may migrate or disappear.
- Signs: **pulse deficit / radio-radial delay**, unequal arm BP, early diastolic murmur (acute AR), pulmonary oedema or shock.
- **Chest pain + stroke** (carotid) or **chest pain + acute limb ischaemia** (iliac) at the same time = suspect dissection.
- Complications: acute AR, MI (usually **RCA**), haemopericardium/tamponade, heart failure, syncope, stroke, paraplegia, mesenteric/renal/limb ischaemia.

Investigations

- **CT aortic angiogram = definitive**; TOE and MRI are alternatives. TTE has inferior accuracy; chest X-ray is not diagnostic.
- ECG: 20% of type A show ischaemia. D-dimer <500 ng/mL may rule out dissection only in low-risk patients.

Management and prognosis

- All patients: ICU, monitoring, arterial line, urinary catheter, IV morphine.
- BP: IV beta-blocker, IV calcium antagonist or IV sodium nitroprusside; **target SBP 100-120 mmHg**.
- **Type A = surgical emergency** (mortality rises 1-2% per hour without surgery); graft replacement.
- **Type B = medical** (tight BP control, serial imaging). Stent or operate if: end-organ ischaemia, ongoing extension (ongoing pain), aneurysmal expansion, Marfan.
- Type A: 40% die in the first 24 h without surgery; 96% alive at 1 year after surgery. Type B: 85% alive at 1 year on medical therapy.

Source: [076] Aortic dissection

PART 1: LECTURE

Chest pain and cardiac presentations [048, 055]

Topic: Cardiac symptoms, presentations and clinical reasoning

History structure

- Introduce yourself, confirm name, get consent. Let the patient tell the story in their own words; avoid leading questions.
- Analyse each complaint: onset, course, duration, aggravating and relieving factors, associated symptoms.
- Cardiac symptoms: chest pain, palpitations, dyspnoea, cough, haemoptysis, leg oedema, syncope.

Chest pain

- First question: is it an emergency? Life-threatening: **coronary ischaemia, aortic dissection, pneumothorax, pulmonary embolism**.
- **Ischaemic:** central/retrosternal/epigastric, crushing, radiates to neck, jaw, left arm; brought on by effort or stress; relieved by rest or nitrates; minutes (longer = infarction); sweating, vomiting.
- **Pericarditis:** localised, stitching, radiates to back; worse on inspiration and lying; **relieved by leaning forward**; malaise.
- **Musculoskeletal/pleuritic:** localised, stitching, worse on breathing, coughing and movement; no radiation.
- Palpitations: time, mode of onset and offset, relation to exertion, duration.

Breathlessness and heart failure

- **Orthopnoea:** needs to sit up; count the pillows. **PND:** wakes about 2 h after sleep and opens the window for air.
- Leg oedema in heart failure builds up during the day; eased by raising the legs.
- **Dyspnoea + cough + leg oedema = heart failure.**
- NYHA: I none; II slight limitation; **III marked limitation, comfortable only at rest**; IV symptoms at rest.

Acute coronary syndrome (055)

- Crushing chest pain + sweating: **ECG first**.
- ACS = unstable angina (troponin -ve), NSTEMI (troponin +ve), STEMI or new LBBB (ST elevation, troponin +ve); all share plaque rupture and similar first treatment.
- Initial drugs: **aspirin 300 mg + clopidogrel 300 mg + enoxaparin 1 mg/kg**.
- Complete heart block after MI (pulse 28): temporary pacing + revascularisation in the cath lab.
- Temporary pacing: asystole, symptomatic bradycardia not responding to atropine, Mobitz II, complete heart block, drug overdose (digoxin, beta-blocker, verapamil). Permanent pacing: persisting symptomatic bradycardia, complete AV block.

Source: [048] Chest pain and cardiac presentations; [055] Cardiopresentations

PART 1: LECTURE

Cardiology history taking and examination [099]

Topic: Cardiac symptoms, presentations and clinical reasoning

History

- Symptom groups: pulmonary congestion (cough, dyspnoea, orthopnoea, PND, haemoptysis), systemic congestion (RUQ pain, oedema, swelling), low output (fatigue, syncope, oliguria), chest pain, palpitations, embolic, constitutional, pressure (dysphagia, hoarseness).
- Risk factors: hypertension, smoking, hyperlipidaemia, diabetes, obesity, family history. **Premature CAD:** patient = man <45, woman <55; family = male <55, female <65.
- Drugs: contraceptive pill raises clot risk; cocaine causes coronary spasm. Finish with ICE and a summary.

JVP

- Patient at **45 degrees**, head turned to the left; use the right IJV (more reliable than EJV, no valves).
- JVP = vertical height above the **sternal angle** (normal up to 3 cm). **CVP = JVP + 5 cm** (normal 8 cm H2O).
- Venous pulsation: 2 waves, inward, not palpable, occludable, falls on inspiration and sitting up, rises with abdominal pressure.
- 'a' wave absent in **AF**; **cannon a** in AV dissociation (complete heart block, VT); large 'a' in pulmonary hypertension or tricuspid stenosis; **large v (systolic expansion) = TR**.
- Raised JVP: right heart failure, tamponade, constriction, RV infarct, fluid overload; **SVC obstruction = non-pulsatile**. Hepatojugular reflux positive if a sustained rise of 4 cm or more.

Precordium and apex

- Normal apex: left 5th space, midclavicular line. Out and down = LV dilatation; diffuse with systolic retraction = RV enlargement.
- **Hyperdynamic** = volume overload (AR, MR, VSD, PDA); **heaving** = pressure overload (AS, coarctation, hypertension); **tapping** = MS.
- Milking test: dilated vein refills from above = SVC obstruction; from below = IVC obstruction.

Auscultation

- Bell (cone) for low-pitched sounds (MS rumble, S3, S4); diaphragm for high-pitched sounds.
- Loud S1 in MS; soft S1 in MR. **Wide fixed split S2 = ASD**; reversed split = LBBB, AS, HCM; narrow split = pulmonary hypertension.
- Metallic click with S1 = mitral valve replacement; with S2 = aortic valve replacement.
- Murmurs: pansystolic (MR, TR, VSD), ejection (AS, PS), early diastolic (AR, PR), mid-diastolic (MS, TS), continuous (PDA). Right-sided murmurs get louder on inspiration; MS and MR in the left lateral position; AR leaning forward in expiration.

Source: [099] History taking; CVS examination; Student notes; Neck veins

PART 1: LECTURE

Clinical cardiac cases S, T, U [039, 040] and Diagnosis unclear [002]

Topic: Cardiac symptoms, presentations and clinical reasoning

Case S: heart failure and dilated cardiomyopathy

- Congestive cardiac failure: dyspnoea, raised JVP, **S3 gallop**, displaced apex, crackles, hepatomegaly, ascites, oedema.
- Causes: coronary disease (two-thirds), hypertension, valve disease, cardiomyopathies, infiltration, high-output states.
- Chest X-ray: cardiomegaly, bat-wing shadowing, upper lobe diversion, septal lines, effusions. **TTE is the first-line imaging test.**
- Acute treatment: sit up, oxygen (SaO2 at least 95%), IV morphine + anti-emetic, **IV furosemide** with or without nitrates, fluid restriction, daily weights.
- Long-term (once LV impairment is confirmed): **ACEi/ARB + beta-blocker + aldosterone antagonist**; second-line hydralazine/ISDN, digoxin; CRT, ICD, LVAD, transplant.
- DCM causes: idiopathic, familial, **alcohol, anthracyclines**, post-myocarditis, peripartum, tachycardia, infections (HIV, Chagas), thiamine/selenium deficiency. Post mortem: nutmeg liver.

Case T: NSTEMI

- Unstable angina = troponin negative; **NSTEMI = troponin positive without ST elevation**; STEMI = ST elevation.
- Immediate: oxygen, morphine, nitrates, **aspirin 300 mg then 75 mg, clopidogrel 300 mg then 75 mg**, anticoagulant. Later: beta-blocker, ACEi, statin, angiography.
- Risk scores: GRACE (in-hospital and 6-month mortality), TIMI, HEART.
- **Ongoing pain or worsening ST depression → urgent inpatient angiography and PCI.**

Case U: infective endocarditis

- Fever + valve lesion + a recent procedure (dental) + splenomegaly + new murmur.
- **Osler's nodes** (tender, finger pulps); **Janeway lesions** (non-tender, palms and soles); splinter haemorrhages; Roth's spots.
- **At least 3 blood culture sets from different sites before antibiotics**; echo. Bloods: anaemia, raised CRP/ESR, low C3.
- New **AV block in aortic valve IE = abscess** reaching the AV node.
- **IV antibiotics for at least 4-6 weeks**; consult surgeons early.

When the diagnosis is unclear

- Symptoms without an organic cause are common and real for the patient.
- Investigate only if the result will change management: tests bring anxiety, harm, radiation (CT chest = 330 chest X-rays), incidental findings and cost.
- Explaining: accept the symptoms are real, give a positive explanation, don't list diagnoses you then exclude, agree a **safety-net**.

Source: [039,040] Cases S, T, U; [002] Diagnosis unclear

PART 2: TOPIC SUMMARY

Arrhythmias and ECG interpretation

Read every ECG in the same order

- Check settings (25 mm/s, 10 mm = 1 mV), then rate, rhythm (find the pacemaker), axis, intervals (PR 120-200 ms, QRS, QTc), chambers, ST-T, Q waves.
- Rate: regular = 300 / large squares between R waves; irregular = QRS in 30 large squares x 10.

Narrow versus wide

Finding	Think of	Unstable	Stable
Irregular, no P	AF	Heparin then synchronised DC	Rate control + anticoagulation
Regular, no P, ~150-200	SVT	Synchronised DC	Adenosine, beta-blocker or verapamil
Wide, regular, AV dissociation	VT (assume VT if structural disease)	Synchronised DC	IV amiodarone
Chaotic	VF	Shock	

AF essentials

- Stroke risk: CHA2DS2-VA 2 or more = anticoagulate; DOACs preferred; warfarin for metallic valve or mitral stenosis.
- LVEF under 40%: no verapamil/diltiazem. Before cardioversion: TOE or 3 weeks of anticoagulation, and 4 weeks after.

Heart block ladder

- 1st degree: long PR, all conducted. Mobitz I: PR lengthens until a drop (AV node, watch). Mobitz II: fixed PR, sudden drop (below the node, pace). Complete: P and QRS independent; broad slow escape = pacemaker.
- Bundle branch: LBBB broad notched R in I, V5-V6; RBBB RSR' in V1. WPW: short PR + delta wave.

Ischaemia and other patterns

- STEMI territories: anterior (LAD) V1-V4; inferior (RCA) II, III, aVF; posterior = tall R and ST depression V1-V3.
- Pericarditis: widespread saddle ST elevation + PR depression. Large effusion: low voltage + electrical alternans.
- Electrolytes: peaked T and broad QRS = high K; U waves = low K; long QT = low K, Ca, Mg.
- LVH: S V1 + R V5 over 35 mm. PE: tachycardia, RBBB, right axis, S1Q3T3.

PART 2: TOPIC SUMMARY

Valvular heart disease and infective endocarditis

Murmur map

Lesion	Murmur	Key signs
Aortic stenosis	Ejection systolic, aortic area to carotids	Slow-rising pulse, soft A2; late peak = severe
Aortic regurgitation	Early diastolic, left sternal edge, sitting forward	Collapsing pulse, displaced apex, Austin Flint
Mitral stenosis	Mid-diastolic rumble, apex, bell, left lateral	Tapping apex, loud S1, opening snap, malar flush, AF
Mitral regurgitation	Pansystolic, apex to axilla	Soft S1, S3; MVP = click + late systolic
Tricuspid regurgitation	Pansystolic, lower left sternal edge, louder on inspiration	Large v waves, pulsatile liver

When to intervene

- AS: severe with symptoms (TAVI if unfit for surgery). AR: symptoms, EF 50% or less, or dilated LV. MS: area 1.5 cm^2 or less with symptoms; balloon only after TOE excludes LA thrombus. MR: symptoms, or ESD over 40 mm or EF under 60%.
- MS with AF: anticoagulate as well as control rate.

Endocarditis: link to valve disease

- Damaged valve + bacteraemia = vegetation. Native valve: viridans strep commonest. Early prosthetic: S. aureus, S. epidermidis. Drug users: tricuspid valve.
- Fever + new murmur + embolic or immune signs (Osler, Janeway, splinters, Roth spots, splenomegaly).
- At least 2-3 blood culture sets before antibiotics, then echo (TTE, then TOE). Definite: 2 major, 1 major + 3 minor, or 5 minor.
- Empirical: native = amoxicillin + gentamicin; prosthetic = vancomycin + gentamicin; at least 4-6 weeks IV.
- Surgery: heart failure, uncontrolled infection (abscess, new heart block), large vegetation or recurrent emboli.

PART 2: TOPIC SUMMARY

Chronic heart failure, cardiomyopathies and pericardial disease

Heart failure core

- Syndrome of breathlessness, ankle swelling and fatigue from structural or functional heart disease. SNS and RAAS harmful; natriuretic peptides protective.
- Classify by EF: reduced 40% or less, mildly reduced 41-49%, preserved 50% or more. NYHA I-IV for symptoms.
- Tests: normal BNP/NT-proBNP in an untreated patient virtually excludes HF; echo is the key test; check iron.
- Prognostic pillars: ACEI/ARB/ARNI + beta-blocker + MRA + SGLT2 inhibitor. Diuretics relieve symptoms only.

Cardiomyopathies at a glance

Type	Key feature	Watch for
HCM	Asymmetric septal hypertrophy, autosomal dominant	Sudden death in the young; murmur louder on standing/ Valsalva; avoid vasodilators
DCM	Dilated, thin-walled, poor systole	Alcohol, anthracyclines, peripartum, myocarditis; treat as HF
ARVC	Fibro-fatty RV	VT, T inversion V1-V3, epsilon wave
Restrictive	Stiff, normal size, big atria	Amyloid: low voltage with thick walls
Tako-tsubo	Apical ballooning after stress	Normal coronaries, recovers in weeks

Pericardium

- Acute pericarditis: sharp pain relieved by leaning forward, rub, saddle ST elevation + PR depression; NSAID + colchicine.
- Tamponade: Beck's triad (high JVP, low BP, quiet heart sounds), pulsus paradoxus over 10 mmHg; echo; urgent pericardiocentesis.
- Constrictive: pericardial knock; TB commonest per the lecture; CT/MRI; differential = restrictive cardiomyopathy.

PART 2: TOPIC SUMMARY

Ischaemic heart disease, dyslipidaemia and hypertension

One disease, three risk factors

- Atherosclerosis drives stable angina (fixed 80-90% stenosis) and ACS (unstable plaque that ruptures, often only 30-50%).
- Modifiable risk: smoking, diabetes, hypertension, dyslipidaemia, inactivity, obesity.

Coronary syndromes

	Troponin	ECG	Plan
Unstable angina	Negative	ST depression, T change or normal	Antiplatelets, anticoagulant, risk score
NSTEMI	Positive	No ST elevation	Same, angiography (urgent if ongoing pain)
STEMI	Positive	ST elevation or new LBBB	Primary PCI (thrombolysis if PCI not possible, within 12 h)

- First: ECG within minutes, aspirin 300 mg, pain relief, oxygen only if SpO2 under 94%.
- Stable angina pathway: ECG, then CT coronary angiography, then stress imaging, then invasive angiography. Treat with aspirin, statin, beta-blocker, nitrate; PCI/CABG for severe ischaemia.
- Arteries: LAD = anterior; RCA = inferior and AV node (90%).

Hypertension (ESC 2024)

- Hypertension 140/90 or more; confirm with home or ambulatory BP. Suspect secondary causes if under 40, resistant or severe.
- Start a single pill of two drugs (ACEi/ARB + CCB or thiazide), then triple, then add spironolactone. Target systolic 120-129.

Lipids

- LDL goals: very high risk under 55 (and 50% fall), high under 70. Steps: maximal statin, add ezetimibe, add PCSK9 antibody.
- Stop statin if ALT over 3x or CK over 10x normal. Familial hypercholesterolaemia: very high LDL, tendon xanthomas, screen family. TG over 500 = pancreatitis risk.

PART 2: TOPIC SUMMARY

Cardiac emergencies (acute heart failure, aortic dissection)

First question for any chest pain: which of the four killers?

- Acute MI, aortic dissection, pulmonary embolism, tension pneumothorax.
- Check pulses and BP in both arms before giving antiplatelets, anticoagulants or thrombolysis: all are contraindicated in dissection.

Acute heart failure

- Breathless, orthopnoea, pink frothy sputum, bilateral crackles, raised JVP.
- Tests: ECG, echo, chest X-ray, ABG, troponin, pro-BNP, renal function.
- Treat: sit up; oxygen if SpO2 under 90%; non-invasive ventilation; IV furosemide (review response at 2-4 h); IV nitrates; inotropes only for hypoperfusion or shock; morphine only for pain.

Aortic dissection

- Tearing pain maximal at onset, to the back; pulse deficit or unequal arm BP; new early diastolic murmur (AR).
- Chest pain together with stroke or a cold limb = think dissection. Can cause inferior MI (RCA) and tamponade.
- CT aortic angiogram is definitive. Lower systolic BP to 100-120 with IV beta-blocker.
- Stanford A (ascending) = emergency surgery. Stanford B = medical, unless organ ischaemia, extension or expansion.

	Acute heart failure	Aortic dissection	STEMI
Key clue	Crackles, high JVP	Pulse deficit	ST elevation
Key test	Echo, BNP	CT aortogram	ECG
First drug	IV furosemide	IV beta-blocker	Aspirin

PART 2: TOPIC SUMMARY

Cardiac symptoms, presentations and clinical reasoning

History

- Let the patient tell the story; analyse each symptom (onset, course, duration, triggers, relief, associated features).
- Ischaemic pain: central, crushing, to neck/jaw/arm, on effort, eased by rest or nitrates. Pericardial: sharp, relieved by leaning forward. Musculoskeletal: localised, worse on movement and pressure.
- Heart failure clues: orthopnoea (count the pillows), PND, evening leg swelling. Premature CAD in family: male under 55, female under 65.

Examination

- JVP at 45 degrees, right internal jugular, height above the sternal angle (normal up to 3 cm). Absent a wave = AF; cannon a = complete heart block; large v = TR; non-pulsatile raised = SVC obstruction.
- Apex: displaced = LV dilatation; heaving = pressure load; hyperdynamic = volume load; tapping = MS.
- Heart sounds: loud S1 in MS; wide fixed split S2 = ASD; reversed split = LBBB or AS; S3 = volume overload; S4 = stiff ventricle.

Case patterns to recognise

- Breathless + raised JVP + S3 + crackles = heart failure; TTE first; ACEi/ARB + beta-blocker + MRA.
- Crushing pain + troponin rise without ST elevation = NSTEMI; aspirin + clopidogrel + anticoagulant; urgent angiography if pain continues.
- Fever + new murmur after dental work = endocarditis; 3 blood culture sets first; new AV block = aortic root abscess.

When the diagnosis is unclear

- Symptoms are real even without an organic cause. Test only if the result changes management.
- Give a positive explanation, avoid listing excluded diseases, and agree a safety net.